

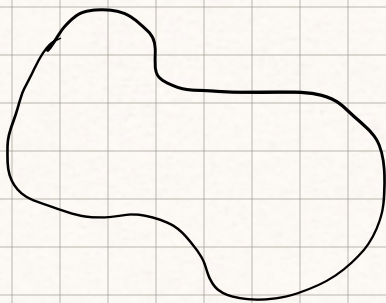
$C(r)$ is a function which computes the circumference of a circle

is $C(2r) = 2C(r)$? yes - its a linear function

$$C(kr) = kC(r)$$

all first order

what if C is the circumference of another shape?



$$\gamma(t) = (x(t), y(t)) \quad t \in [a, b]$$

length is $\int_a^b \sqrt{(x'(t))^2 + (y'(t))^2} dt$

the length of 2γ is $(2x(t), 2y(t))$

$$\text{is } \int_a^b \sqrt{(2x')^2 + (2y')^2} dt = 2 \int_a^b \sqrt{(x')^2 + (y')^2} dt$$

Circumference of any curve scales w/ dilation

do	don't
circumference	$y = mx + b$
$f(x) = x^n \quad n \neq 1, 0$	$f(x) = x^n \quad n \neq 1, 0$

What about $f(x) = \ln x$ $f(kx) = \ln(kx)$ works

What about trig?

$$\sin\left(2\frac{\pi}{4}\right) = 2\sin\left(\frac{\pi}{4}\right) \text{ no}$$

property 1 $f(kx) = kf(x)$

do	don't
0th and 1st order polynomials	trig e^x $y = mx + b$

all "do" also satisfy

$$f(x_1 + x_2) = f(x_1) + f(x_2) \leftarrow \text{property 2}$$

Consider the function $D: \text{functions} \rightarrow \text{functions}$

$$D(f(x)) = f'(x) \quad D \text{ has properties 1 and 2}$$

Any function satisfying properties 1 and 2 is linear

$f(x) = mx + b$ is only linear when $b = 0$

$$\text{fix } a, b \quad I(f(x)) = \int_a^b f(x) dx$$

this is linear $I: \text{functions} \rightarrow \mathbb{R}$

Def: let P_n be the set of polynomials degree n

note: $D: P_n \rightarrow P_{n-1} \quad n \geq 1$